# Kaiyuan Liu

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## Education

**B.S.** **University of Washington, Seattle** 2022 ~2026
**Double Major:** Computer Science, Mathematics

- GPA: 3.93/4.0; Honor Math Sequence; Deans List
- Departmental Honors in Computer Science

## Awards

**ICPC World Final Honor**, ICPC Foundation 2024.09
**CRA Undergraduate Research Award Honorable Mention**, CRA 2025.09
**Shenoy Undergraduate Research Fellowship**, Simons Foundation 2024 ~2025
**CSE Award for Excellence Scholarship**, University of Washington 2024 ~2025
**ICPC North American Championship 12th Place**, ICPC Foundation 2024.05
**UW Winter Programming Contest Champion**, University of Washington 2023 ~2024
**Tinker Research Grant**, 5000$ credit, Thinking Machines Lab 2025.11

## Publications

**LiveCodeBench Pro: How Do Olympiad Medalists Judge LLMs in Competitive Programming?**
Zihan Zheng*, Zerui Cheng*, Zeyu Shen*, Shang Zhou*, **Kaiyuan Liu***, *et al.* *Equal contribution
*The Thirty-Ninth Annual Conference on Neural Information Processing Systems (NeurIPS 2025)*
arxiv.org/abs/2506.11928

**AutoCode: LLMs as Problem Setters for Competitive Programming**
Shang Zhou*, Zihan Zheng*, **Kaiyuan Liu***, *et al.* *Equal contribution
arxiv.org/abs/2510.12803
The Fourteenth International Conference on Learning Representations (ICLR 2026)

**Evaluating LLM Agents as Human Simulators in Climate Social Dilemmas**
**Kaiyuan Liu***, Xiaoxuan Hou*, Jiayi Yuan, Natasha Jaques *Equal contribution
Preview Link
Under Review at *ACL 2026*

**FrontierCS: Evolving Challenges for Evolving Intelligence**
Qiuyang Mang*, Wenhao Chai*, Zhifei Li*, Huanzhi Mao*, Shang Zhou*, *et al.* *Equal contribution
arxiv.org/abs/2512.15699
Under Review at *ICML 2026*

## Experience

**Social RL Lab**, Research Assistant — Seattle, WA, 2025.01 ~Present

- Working on various projects related to multi-agent, reinforcement learning, and large language models.
- Mentored by Dr. Natasha Jaques.

**Allen Institute**, Research Assistant — Seattle, WA, 2024.09 ~2025.05

- Applied Deep Learning and Reinforcement Learning to analyze representations from artificial models and neural data from Multi-Armed Bandit mouse foraging tasks. Explored brain representations and dynamics in decision making

**VecML**, Machine Learning Engineer — Seattle, WA, 2024.06 ~2024.09

- VecML is a startup company focusing on Machine Learning System Infrastructure.
- Tested and developed vector databases for Retrieval-Augmented Generation.
- Designed and implemented an Memory-Disk Hybrid Architecture for fast Top-K Nearest Neighborhood Search

**University of Washington**, Teaching Assistant — Seattle, WA, 2023 ~2024

- Assisted in undergraduate algorithm courses for both major (CSE 421) and non-major (CSE 417) students.
- Served as leading TA in CSE 421 2024 Winter.

## Professional Service

**Reviewer**, NeurIPS Workshop 2025, AAAI 2026, AAMAS 2026
**ICPC Coach**, University of Washington ICPC 2026

## Projects

**Neural Translator: Predicting Neural Activity from Other Brain Regions** — Comp Neuro, NLP

- Developed a neural translator model that predicts neural activity in one brain region from activity patterns in other regions using deep learning and sequence-modeling techniques.

**Neural Spike Vectorization with Attention** — RL, NLP, Comp Neuro

- Developed an attention-based model for vectorizing brain spike sequences.
- Compared latent representations of RL-trained RNNs with vectorized neural activity from mice performing decision-making tasks using optimal-transport metrics.

**Fine-grained Chinese Toxic Language Detection** — NLP

- Reimplemented and improved Chinese toxic language classifier based on BERT
- Authored technical poster and report

**Multi-Agent Reinforcement Learning Survey** — RL

- Conducted literature review on MARL algorithms, especially in games.

**Variational Autoencoder for Neural Data** — DL

- Implemented VAE model for analyzing high-dimensional neural data

## Technologies

**Languages:** Python, PyTorch, JAX, C++, Java, TypeScript, LaTeX

**Databases:** MySQL, NoSQL, PostgreSQL, SQLite, Azure

**Topics:** Algorithm Design, Reinforcement Learning, Machine Learning, NLP, Computational Neuroscience

**Languages:** English, Chinese (Mandarin)